Multi-Class Intrusion Detection and Prediction using Deep Artificial Neural Networks

Md. Ismail Hossain Shezan
ID: 20234103389
Department of Computer Science and Engineering
Bangladesh University of Business and Technology
Email: ismailshezan070@gmail.com

Md Jobaer Islam Alif
ID:20234103391
Department of Computer Science and Engineering
Bangladesh University of Business and Technology
Email: alifjobaer12@gmail.com

ABSTRACT

Cybersecurity attacks are exponentially increasing, making existing detection mechanisms insufficient and enhancing the necessity to design more relevant prediction models and approaches. This issue is still an open research problem since existing attack prediction models are failing to follow the huge number of attacks and their variety. Recently, machine learning approaches and especially deep learning [9] techniques have received much attention from researchers since their unparalleled high performance in several prediction-based fields. In this context, this paper explores the application of deep learning techniques for predicting cybersecurity attacks. Particularly, it proposes a new **ANN** (Artificial Neural Network), and **MLP** (Multilayer Perceptron) based models carefully designed to predict the type of attack potentially to happen. The proposed models were validated using an available dataset called Intrusion Detection Evaluation Dataset (CIC-IDS2017) [1]. showing encouraging results especially for the ANN model with an accuracy of 94.78%. This work demonstrates the significance of Deep learning [9] in Cyber-Security.

KEYWORDS

Attack prediction, deep learning techniques, cybersecurity

INTRODUCTION

Cyberattacks, which represent a ubiquitous threat to organizations, enterprises, governments, and countries, are increasing in volume. Everyday with the growth of technology the misuse of that tech is also increasing rapidly. The attack is also getting advanced every day. According to the Norton Security report of the first half of 2019, the number of exposed records is 4.1 billion, and the ratio of reported breaches has increased by more than 54% compared to the first half of 2018. These numbers give an idea about the exponentially increasing threats for organizations.

With the exponential increase of attack the pattern of each attach is also changing rapidly. People are adapting the prediction detection systems. Before people manually checked for each attack. But with the advance pace of technology and AI people can’t keep track of everything. The traditional model that are used are also getting old. They simply can’t keep up with the new pattern of attack that are happening now and then.

Therefore, there is a great demand for attack prediction tools that can inform administrators of future attacks and malicious activities. This way, administrators could react to these upcoming attacks and deploy prevention mechanisms before causing harm to their organizations. For this reason, security administrators

usually seek a viable attack prediction tool that helps in predicting attacks before these attacks take place. Although research effort on attack prediction have reached a higher state. It is still challenging to detect with high accuracy. In this paper we have try to tackle that problem. We proposed a deep learning method that have a higher f1-score and a good accuracy that not only tell the occurrence of the attack but it's class. More precisely the contribution of this paper can be summarized as follow:

- A Deep learning model that predicts the occurrence of a future attack.
- The model used a dataset that has a higher amount of data and a variety of class on different type of attack pattern.
- The dataset consists large scale network traffic analysis to have a stable and consistence result
- Evaluation result that shows a higher score of f-measure of the model.

This model has a high dimensional state that consist of 78 input features. For a human analyst, identifying subtle correlations and malicious patterns across such a vast number of variables is nearly impossible. But this deep learning model can excel in those environments. By processing these 78 fields simultaneously through its hidden layers, the model can uncover intricate and 'tricky' attack patterns that would otherwise remain hidden. Among the dataset there consists of some noise. This model is designed to carefully handle that irregularity and give the correct prediction with exceptional accuracy.

The rest of this paper is organized as follows: Section 2 provides a review of related literature and previous works in attack prediction. Section 3 describes the materials and methods, including dataset characteristics and the proposed deep learning architecture. Section 4 presents the experimental results and discussion. Finally, Section 5 concludes the paper and suggests future research directions.

LITERATURE REVIEW

In the early research of cybersecurity people focused on **Signature based detection** system. Problem with those legacy frameworks is that they operated by matching the existing known attack pattern. While it's effective determining the previously documented attacks it struggles with newly develop pattern. As we all know with new day the complexity of attack pattern is also increasing. This system simply can not keep up with the modern world. This limitation sparked a shift toward **Anomaly-based detection**, which seeks to identify variations from "normal" network behavior rather than relying on a static blacklist.

With the rise of large data researchers have started to implement simple **ML** (Machine Learning) algorithm to detect the upcoming threat. Now the convenience side of this is this model can learn from previous attack. Algorithm such as **SVM (**support vector machine), **RF** (Random Forest), **K-NN** (K Nearest Neighbor) have left a huge leap in the history of cybersecurity. However, they also felt a bottleneck with the coming of new age. This model is struggling with high dimensional data such as 78 features set that is found in the modern networking traffic.

Recent advancement has shifted toward ANN especially Deep learning. Unlike ML, deep learning can extract information from raw data. It can easily handle data with high dimensional such as a input layer that consists 78 features. However, many of these models are either highly complex utilizing hybrid CNN (Convolutional Neural Network) LSTM (Long Short-term Memory) structures that require massive computational power or are optimized only for binary (Attack vs. Normal) classification.

Comparison of Previous Study on Related topic:

Study	Model Type	Accuracy	Focus key
Mujeeb At al. [2]	CNN/LSTM	96.2%	Focused on temporal pattern
ResearchGate [8]	CNN/LSTM	95-99%	Captured both time and space correlation
Jairu [3]	Decision Tree	96.37%	Fast processing with 14 selected features.
Bangari [4]	Random Forest	98.38%	Best performance among "shallow" ML models.
Proposed Model	3-Layer ANN	94.78%	Multiclass (15 types) & 78-feature efficiency.

Table-1: comparison between different study's

Although other modal has higher accuracy compare to proposed model, they are relatively slow. And can't handle a large amount of different type of attack. In our model the prediction time is fast pace and handle 78 feature input layers. By maintaining a streamline our model is designed to act fast with a approximately 95 f1-score that is suitable for real life network environments. This paper aims to fill that gap by proposing a 3-layer ANN architecture using the keras framework [6] that balances computational speed with high-precision multi-class detection.

METERIALS AND METHODS

System Workflow:

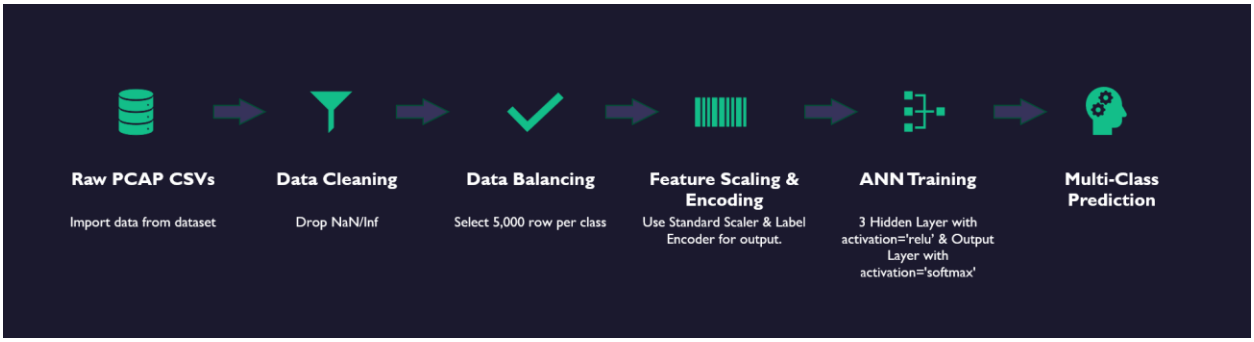

Fig-1: a descriptive picture of system workflow

The model has first taken 8 different type of dataset (.csv file) that has combined into one. Then the raw data is first cleaned since there is a lot of inconsistency among them. To balance those data and bring consistency to it. The system first dropped those null values; we have selected around 5000 rows approximately for each class. There is around 15 different types of distinct class of cyber-attack. Also, each row consists of 79 columns. Then those selected data have been fine tuned for the model to use its value. We have use standard scaler and label encoder from the Scikit-learn library [7] to feature tuning the data.

When the data is preprocessed, we have trained our 3-layer ANN model [6]. Now it can be used for multi class attack prediction.

Dataset:

For the model we have used "Intrusion Detection Evaluation Dataset (CIC-IDS2017) [1]" dataset that consist around 2.8 million data. It has a input layer of 78 feature. The dataset has around 15 type of attack class that includes DDOS, Botnet etc. Here is a sample pic of dataset.

Destination Port	Flow Duration	Total Fwd Packets	Total Backward Paci	Total Length of Fwd	Total Length of Bwd	Fwd Packet Length M	Fwd Packet Length M	Fwd Packet Length M	Fwd Packet Length S	Bwd Packet Length M	Bwd Packet Le
443	29962393	10	11	685	5645	333	0	68.5	113.606778	1821	
80	107	2	0	12	0	6	6	6	0	0	
53	2688125	2	2	68	180	34	34	34	0	90	
53	37102226	2	2	127	345	81	46	63.5	24.74873734	232	
443	16754213	12	12	751	7039	436	0	62.58333333	132.9103993	1610	
443	116717561	22	18	1179	3226	597	0	53.59090909	146.054021	1448	
1186	75	1	1	0	6	0	0	0	0	6	
1011	56	1	1	2	6	2	2	2	0	6	
53	30588	4	2	164	170	41	41	41	0	85	
22051	8230042	1	5	1375	24	1375	1375	1375	0	6	

Fig-2: dataset of 79 column of network traffic

Class Labels	Number of instances
BENIGN	2359087
DoS Hulk	231072
PortScan	158930
DDoS	41835
DoS GoldenEye	10293
FTP-Patator	7938
SSH-Patator	5897
DoS slowloris	5796
DoS Slowhttptest	5499
Botnet	1966
Web Attack – Brute Force	1507
Web Attack – XSS	652
Infiltration	36
Web Attack – Sql Injection	21
Heartbleed	11

Fig-3: classed based instance occurrences

In the dataset there are some inconsistence data. So, we have removed that have null data or false data and selected around 5000 rows for each class. Some classes that do not have that many rows for them we have created some demo data to reached a decent accuracy. And classes that have more than 5000 row we have selected row randomly and bring the scale to 5000 for easier data handle and less false data prediction. Here is a fig of data after preprocessing them for the model.

Destination Port	Flow Duration	Total Fwd Packets	Total Backward Paci	Total Length of Fwd	Total Length of Bwd	Fwd Packet Length M	Fwd Packet Length M	Fwd Packet Length M	Fwd Packet Length S	Bwd Packet Length M	Bwd Packet Le
80	98711033	7	7	327	11595	321	0	46.71428571	120.9692995	5792	
80	11866645	5	4	439	3525	439	0	87.8	196.3267684	3525	
80	1782435	3	6	26	11601	20	0	8.666666667	10.26320288	2920	
1501	43	1	1	2	6	2	2	2	0	6	
53	1029954	2	2	109	253	58	51	54.5	4.949747468	186	
80	5350092	3	1	0	0	0	0	0	0	0	
51695	41	1	1	6	6	6	6	6	0	6	
52314	34	1	1	6	6	6	6	6	0	6	
80	92284681	8	4	56	11601	20	0	7	5.656854249	7215	
80	64	1	1	0	0	0	0	0	0	0	

Fig-4: data of dataset after preprocessing them

We have preprocessed these data by using level encoding. We have turned string level such as DDoS, Botnet, Dos Hulk into a string of array [0,1,2] using level encoding. The Processed data has been split with the ratio of 0.2. and train over 1000 epochs with batch size of 64. Now to actually train the model first the data need to be feature tuned. So used a standard scale to featured tuned to bring the value to a level for the model to actually work. Here is an image of the table after it's featured scaled:

0	1	2	3	4	5	6	7	8	9	10	
-0.2562439534	1.708020554	-0.278702065	-0.2839136299	-0.1087209972	-0.2610089639	-0.2109518206	-0.09361478979	0.5206135494	0.09112269231	-0.4818040943	-0.182144
-0.2170819696	2.096597739	3.334482731	2.995974734	-0.06050450633	3.941089909	2.78983046	-0.09361478979	-0.2697810955	0.01609840698	3.321525246	-0.182144
-0.2562439534	-0.6873973899	-0.2955558154	-0.2870887396	-0.1213218512	-0.2610121636	-0.3794859093	-0.09361478979	-0.289319612	-0.3926138407	-0.4832373298	-0.182144
-0.2591488258	-0.6855714589	-0.294259373	-0.2855011847	-0.1209350481	-0.2607529832	-0.3510321021	0.2691190747	-0.1152843872	-0.3926138407	-0.425191291	9.98767
-0.2562439534	-0.6858367182	-0.2929629307	-0.2807385201	-0.1211929169	-0.2548222308	-0.3648942133	-0.09361478979	-0.2506451176	-0.3421409073	1.24022839	-0.182144
-0.2562439534	1.237121048	-0.286480719	-0.2807385201	-0.1210441464	-0.2548222308	-0.3648942133	-0.09361478979	-0.2580825203	-0.3647942574	1.937736346	-0.182144
-0.2562439534	0.786917325	-0.2877771613	-0.2886762945	-0.1213218512	-0.2610121636	-0.3794859093	-0.09361478979	-0.289319612	-0.3926138407	-0.4832373298	-0.182144
0.6044589865	-0.6859030622	-0.2916664884	-0.2839136299	-0.1202953354	-0.2609407024	-0.2372168734	-0.09361478979	-0.05838825603	0.07724715505	-0.4526616386	-0.182144
-0.2562439534	-0.5547026651	-0.2929629307	-0.2870887396	-0.1213218512	-0.2610121636	-0.3794859093	-0.09361478979	-0.289319612	-0.3926138407	-0.4832373298	-0.182144
-0.2170819696	2.097475011	3.329296961	2.895958777	-0.05332385439	3.9403177	3.84626925	-0.09361478979	-0.2674429584	0.1497168079	2.629750232	-0.182144

Fig-5: image of dataset after feature tuning

Proposed Methodology:

This model works on a 3 layer ANN system. Where it under goes 3 different dense layer and activation function to reach the output layer. The Dense layers facilitate feature extraction through a linear transformation followed by a non-linear activation. Each neuron calculates the weighted sum of its inputs, adds a bias term, and applies the RELU activation function, defined as:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + bias\right) \tag{1}$$

Here.

W represent weight, f is activation function, x is input.

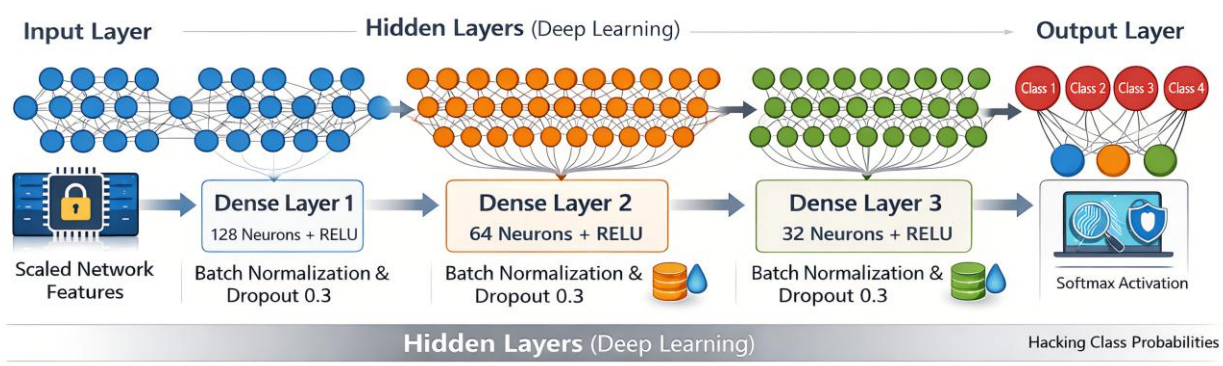

Fig-6: An image of the model methodology

This allows the model to first map the 78 input features into a higher-dimensional space of 128 neurons to intricate the correlation necessary for multiclass classification. After that the data goes through another dense layer that has 64 neurons this time and uses RELU as its activation function. After calculation it again goes through a 3rd dense layer for calculation and again with RELU as its activation function. And in the end the output result uses a SoftMax Activation function to show the predicted result.

RESULTS AND DISCUSSION:

The performance of the proposed model is evaluated using a group of matrices to ensure reliability over all 15 distinct attack class. As illustrated in the Accuracy and loss graph the model demonstrated a overall result of 94.78% while the loss function is consistently minimized to validate the 128-64-32 ANN model.

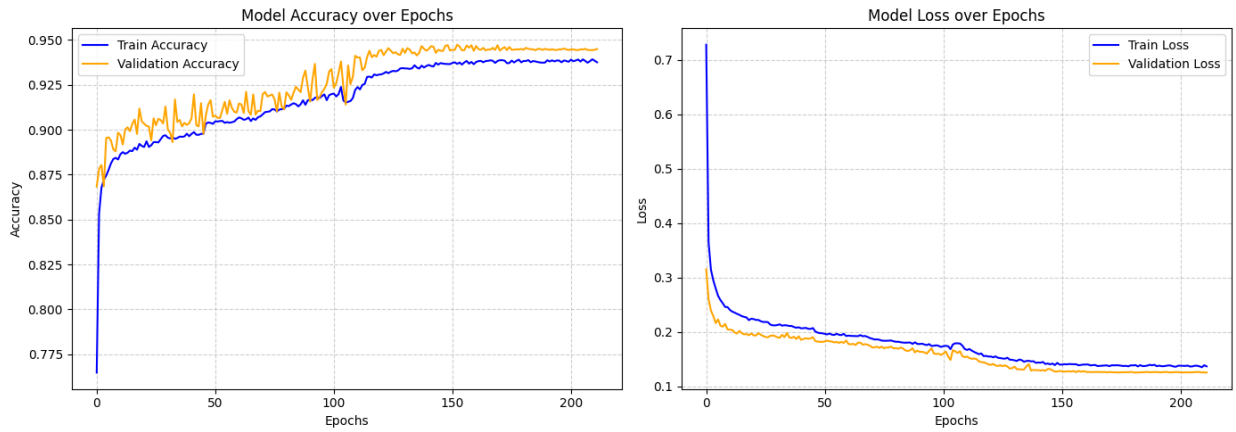

Fig-7: Model Accuracy and Loss over Epochs

To farther analyze the model's ability to distinguished between attack a confusion matrix has been generated.

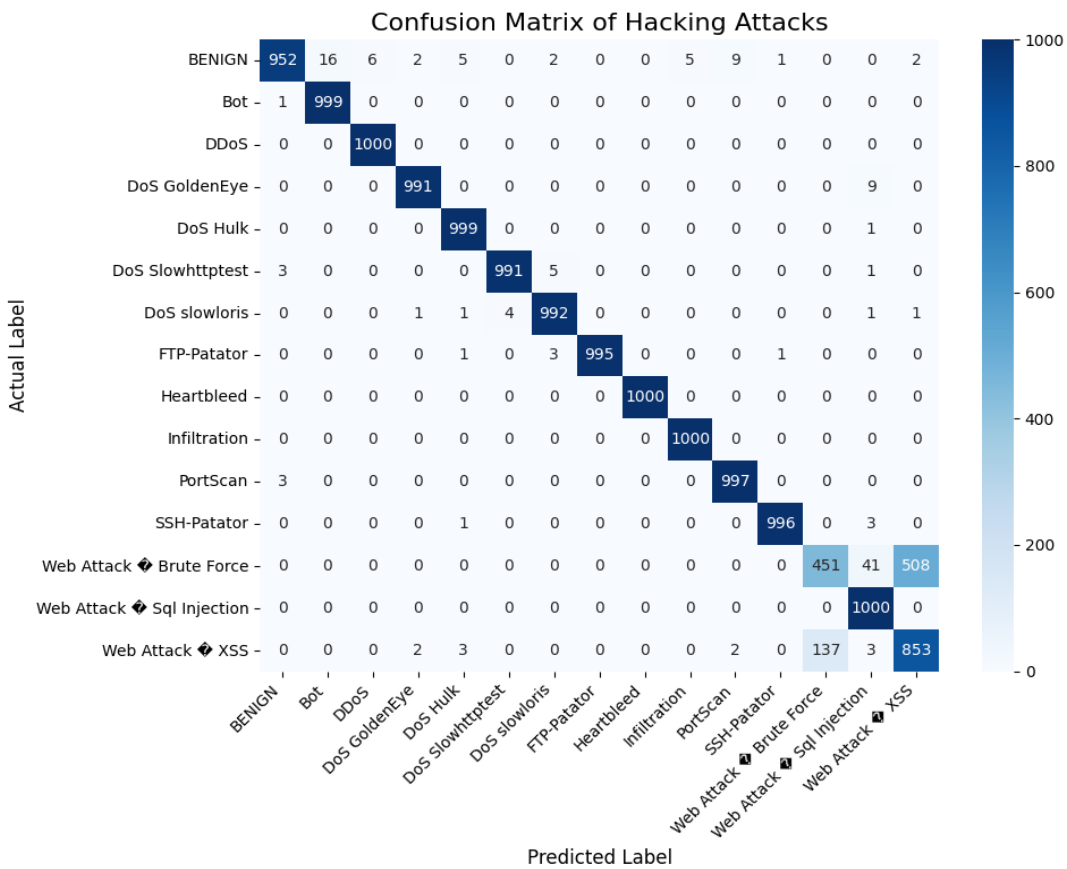

Fig-8: confusion matrix

This matrix reveals that it holds a high precision even with a 15 distinct different category of attacks.

Several statical matrix has been derived from the confusion matrix. This formulas precision, accuracy, recall, f1-score give a broad view of the project. The mathematical form of this matrix's are

$$Accuracy = (\frac{TP + TN}{TP + TN + FP + FN}) \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$F1\ Score = 2 * \frac{Precision\ .\ Recall}{Precision + Recall} \tag{5}$$

Where,

TP = True Positive; when the predicted value is Positive and the actual value is also Positive

TN= True Negative; when the predicted value is Negative and the actual value is also Negative

FP = False Positive; when the predicted value is positive but the actual value is Negative

FN = False Negative; when the predicted value is Negative but the actual value is Positive

Here is a more demonstrative report about the model using these formulas to better understand the performance of this model. In the report we have shown the actual performance in regard to each classification of these attacks.

Class	Precision	Recall	F1 Score	Support
1	0.99	0.95	0.97	1000
2	0.99	1.00	0.99	1000
3	0.99	1.00	1.00	1000
4	0.99	1.00	1.00	1000
5	0.99	1.00	0.99	1000
6	0.99	0.99	0.99	1000
7	0.99	0.99	0.99	1000
8	1.00	0.99	1.00	1000
9	1.00	1.00	1.00	1000
10	1.00	1.00	1.00	1000
11	0.99	1.00	0.99	1000
12	1.00	1.00	1.00	1000
13	0.75	0.48	0.59	1000
14	0.95	1.00	0.97	1000
15	0.63	0.82	0.72	1000

Table-2: Classification Report

From the table we can deduct that,

Accuracy			0.95	15000
Macro avg	0.95	0.95	0.95	15000
Weighted avg	0.95	0.95	0.95	15000

In the report we can see that some class have low values, the reason is that those attacks are relatively low and the data about them is little less in the world. These attacks are extremely tricky to detect since they normally blend its behaviors with the user. The discussion of the result is that even thought we can detect almost all the attack perfectly some of them need a special engineering or layer for better prediction. With the upcoming new data about this attack the prediction will only grow its ability each day.

The stability of these attacks is further shown with a **ROC** curve

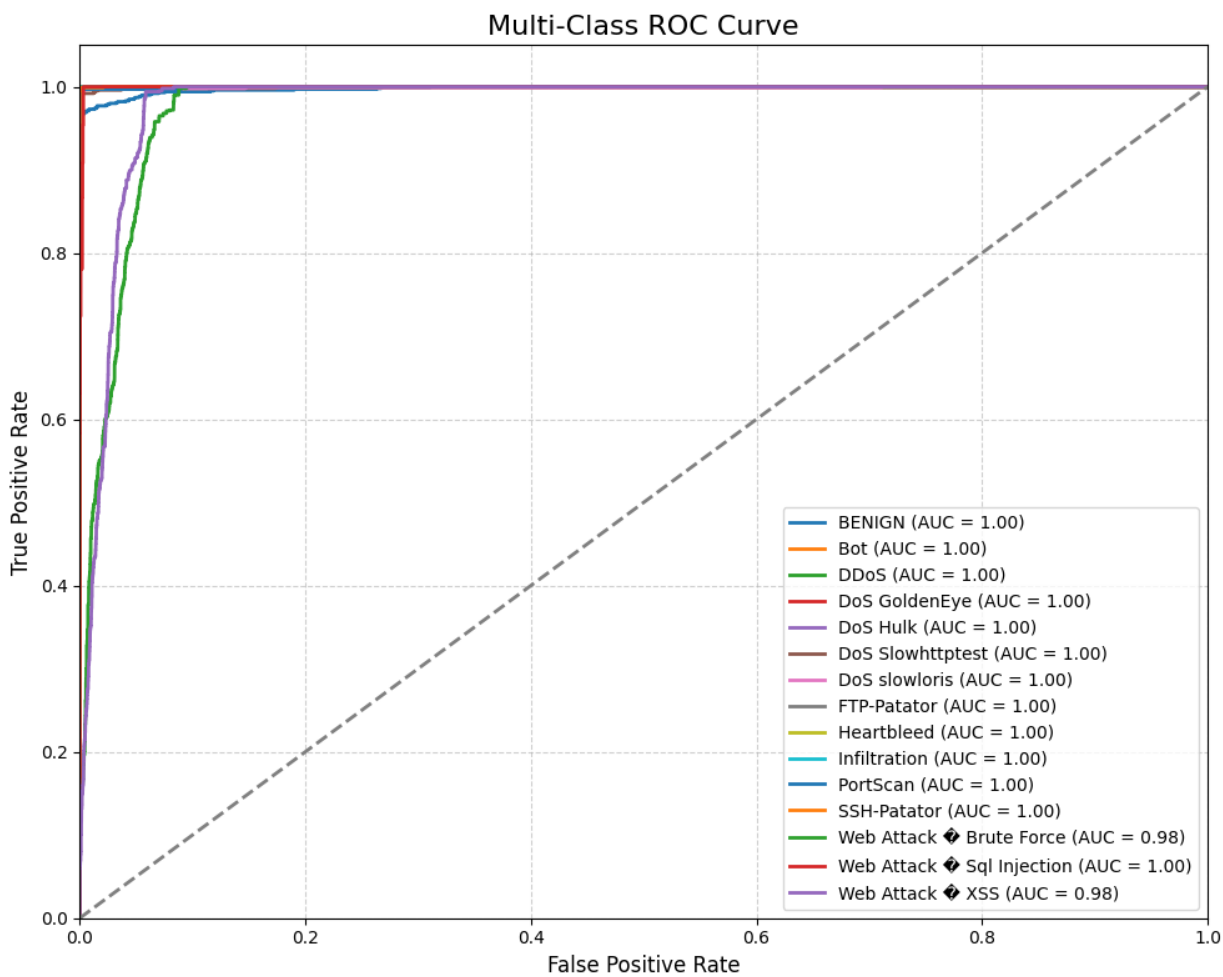

Fig-9: ROC curve

With all these data and analysis in can be conclude that the model is working with great performance. Even though some of the previous studies have a little higher accuracy as shown in the comparison table (table-1) model those were actually worked with 2 to 10 different classes. Also, they are extremely slow for the real-world experience. Even though our model has a little lower accuracy rate its works better for real word environment that work fast and accurate.

CONCLUSION

This research successfully developed the multi class intrusion detection and prediction system. That is a 3-layer ANN [6] model integrating one of the latest datasets called CICIDS2017 [1] whose data was recorded over a week of analyzed network traffic. The system utilizes 78 features for input layer to detect or predict an upcoming attack. The model has addressed a difficult challenge of detecting a 15 distinct type of intrusions within modern network traffic. The model has achieved and macro F1-score of 95. By utilizing the 3-layer deep learning architecture 128-64-32, the model has shown an accuracy around 94.78%.

Future research will work on for the model to verify on more different type of attack and so more to polish the prediction ability to a higher level. The project might integrate newer hybrid ANN model to bring its efficiency to a greater level. Ultimately this tool can act as a framework for automated threat detection becoming an essential tool secures the network infostructures.

REFERENCES

[1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward a reliable intrusion detection benchmark dataset," in *2018 7th International Conference on Software Networking (ICSN)*, IEEE, 2018, pp. 24–31.

[2] M. Mujeeb, R. Ahmed, and S. Ullah, "Deep Learning for Intrusion Detection: A CNN-LSTM Approach," *Procedia Computer Science*, vol. 151, pp. 12-18, 2019.

[3] R. Jairu, "Optimized Decision Trees for Network Security," *International Journal of Computer Science and Network Security*, vol. 8, no. 2, 2021.

[4] S. Bangari and P. Kumar, "Evaluation of Random Forest Performance in Multiclass IDS," *Journal of Cybersecurity Research*, vol. 4, pp. 88-102, 2022.

[5] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous systems," 2015.

[6] F. Chollet et al., "Keras," 2015.

[7] P. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.

[8] ResearchGate, "Network Intrusion Detection using CNN and LSTM for High Dimensional Data," *ResearchGate Technical Report*, 2022.

[9] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.